

Internship Report on
"FULLSTACK WEB DEVELOPMENT"

Submitted to



RANI CHANNAMMA UNIVERSITY BELAGAVI

PeopleTree Education Society, Belagavi

In partial fulfillment of the requirements for the award of Degree in

Bachelor of Computer Application

By

Mr. Vijayanand Rathod

UUCMS: U15BM21S0063



PeopleTree Education Society's

College of Computer Application

CTS No10588/1,2,3. Behind Ramdev hotel

Nehru Nagar Belagavi-590010

PeopleTree Education Society's

College of Bachelor of Computer of Application



Certificate

Certified that the Internship entitled “ **FULL STACK WEB DEVELOPMENT** “ is a Bonafide work carried out by **Mr. Vijayanand Rathod** (U15BM21S0063) in partial fulfilment for the award of Bachelor of Computer Application of Rani Channamma University Belagavi, during the year 2023-2024. The internship report has been approved as it satisfies the academic requirements in respect of Internship work prescribed for the said Bachelor of Computer Application Degree.

Prof. Manoj Narayankar

Internship mentor

Prof. Anil Halabhavi

HOD of BCA department

Prof. Abhisheka Jhanavi

Internship Coordinator

Prof. Avinash Akki

Principal

Name of the Evaluators

Signature with Date

ABSTRACT

This internship report delves into the multifaceted domain of Full stack web development, chronicling the practical experiences and technical proficiencies garnered during the internship tenure. Full stack development embodies a comprehensive approach to web development, encompassing both frontend and backend technologies to create cohesive, dynamic, and responsive web applications. The report elucidates the various stages of the development lifecycle, from requirement analysis and design to implementation, testing, and deployment. During the internship, significant emphasis was placed on mastering key technologies such as HTML, CSS, JavaScript, and modern frontend frameworks like React.js for developing user interfaces. Concurrently, backend development was addressed using Node.js and Express.js, with MongoDB as the database solution. This amalgamation of technologies facilitated the creation of scalable and maintainable applications, illustrating the critical role of a Full stack developer in bridging the gap between client-side aesthetics and server-side functionality. Moreover, the internship underscored the importance of version control systems like Git, collaborative tools, and agile methodologies in streamlining the development process and enhancing team productivity. Practical challenges encountered, such as debugging complex codebases and optimizing application performance, provided invaluable learning experiences and reinforced problem-solving skills. In conclusion, the internship offered a profound understanding of Full stack web development, equipping the intern with a robust skill set and practical insights essential for a successful career in the ever-evolving tech industry. This report encapsulates the journey of integrating theoretical knowledge with hands-on practice, ultimately contributing to the intern's professional growth and readiness for future in web development.

ACKNOWLEDGEMENT

It is my proud privilege and duty to acknowledge the kind help and guidance receive from several people in preparation of this report. It would not have been possible to prepare this report in this form without valuable suggestions, cooperation and guidance.

I would like to express my special appreciation and thanks to my internship guide Prof. Sushma Hiremath Department of Computer Applications, PeopleTree Education Society, Belagavi for excellent guidance and patience. I have benefited greatly from her wealth of knowledge and meticulous editing.

My sincere gratitude to Prof. Shubha Patil

DECLARATION

I **Vijayanand Rathod** bearing **Reg No: U15BM21S0063**, student of Bachelor of Computer Application Rani Channamma University Belagavi ,hereby declare that the internship work entitled “**Full Stack Web Development**” submitted by me, for the award of the Bachelor’s degree in Computer Application to RCU University is a record of bonafide work carried out independently by me. I further declare that the work reported in this internship work has not been submitted And will not be submitted, either in part or in full, for the award of any other degree in this university or any other institute or University.

Place: Belagavi

Name: Vijayanand Rathod

Date:

UUCMS: U15BM21S0063



Accredited by NASSCOM, Approved by the Government of India

Certificate of Internship

This is to certify that

Vijayanand Rathod

has successfully completed internship
From 24th April to 31st May 2024

Full Stack Web Development Internship Project

and has shown competency by delivering the same on/within the
stipulated time has delivered quality work in the project



A handwritten signature in black ink, appearing to read "A.S. Kalpathi".

Abishek Kalpathi
(Business Head, Edureka Learning Center)

1. Introduction

- Overview of the Internship Program
- Company Profile: Edureka
- Internship Objectives
- Scope of the Report

2. Internship Methodology

- Internship Program Structure
- Learning Process
- Tools and Technologies Used
- Mentorship and Guidance

3. Technical Skills Acquired

- XAMPP
- Coding
- Page Development
- React JS
- SQL

4. Projects and Assignments

- Detailed Descriptions of Projects
- Problem Statements, Solution, Approaches
- Role and Contributions
- Screenshots and code Snippets

5. Challenges and Learnings

- Challenges Faced and Overcoming Strategies

- o Key Learnings and Insights

o Importance of Teamwork and Collaboration

6. Certificates

- o Full stack web development

- o React JS

7. Conclusion

- o Summary of Internship Experience
- o Acquired Skills and Future Career Goals
- o Gratitude

1. Introduction

Overview of the Internship Program

My internship at Edureka was a comprehensive and enriching experience that spanned [duration of internship, e.g., three months], focusing on Full Stack Web Development. The program was structured to provide both theoretical knowledge and practical skills necessary for a successful career in web development. The internship included various modules, assignments, and projects designed to enhance my understanding of modern web technologies.

Company Profile: Edureka

Edureka is a leading online education provider specializing in training individuals in the latest technologies. Founded with the mission to provide quality education accessible to everyone, Edureka offers a wide range of courses in fields such as data science, cloud computing, and web development. The company's innovative approach to education, combining expert instructors with real-world projects, ensures that students gain practical skills that are directly applicable in the job market.

Edureka Learning Center is a division of Edureka, a prominent figure in the online professional and higher education sphere specializing in emerging technologies and management domains. Established in 2011, Edureka pioneered the concept of instructor-led live online training within India, guided by a commitment to enhancing students' skill sets.

In September 2021, Edureka became a part of Veranda, a publicly listed company, aligning with its vision of creating a comprehensive education ecosystem catering to diverse learners across India.

Our Vision

To become the largest online learning ecosystem for continuing education, in partnership with corporates and academia.

Our Mission

To create an alternate platform for students who wish to continue and complete courses by attending live online courses, using a team of ridiculously committed educators who will stop at nothing to impart education, helped by a 24 x 7 support system. By deploying our world class team of industry experts, we aim to educate our learners with the skills they need to advance their professional life to the next level.

Internship Objectives

The primary objectives of the internship were to:

- Gain hands-on experience with Full Stack Web Development.
- Develop proficiency in using tools and technologies such as XAMPP, React JS, and SQL.
- Learn to design and develop robust web applications.
- Understand the integration of front-end and back-end components.
- Enhance problem-solving skills through real-world projects.

Scope of the Report

This report provides a detailed account of my internship experience at Edureka. It covers the following areas:

- **XAMPP:** Role in web development, setup process, server configuration, and database management.
- **Coding:** Overview of programming languages and concepts learned (HTML, CSS, JavaScript, PHP).
- **Login Page Development:** Process of designing and developing a login page, integration of front-end and back-end components.
- **React JS:** Fundamentals of React JS, including components, state management, and JSX.
- **SQL:** Concepts learned (data types, queries, joins, etc.) and interaction with databases.

2. INTERNSHIP METHODOLOGY

Internship Program Structure

The internship was structured in a hybrid format, combining online lectures and practical sessions. This format allowed flexibility in learning while ensuring that all participants had access to the necessary resources and support. Weekly sessions included live lectures, recorded tutorials, and interactive Q&A sessions with instructors.

Learning Process

The learning process at Edureka was divided into several phases:

- **Lectures:** Comprehensive lectures covering theoretical aspects of web development.
- **Assignments:** Regular assignments to apply the concepts learned in lectures.
- **Projects:** Real-world projects that required integrating multiple skills and technologies.
- **Practical Exercises:** Hands-on exercises to reinforce learning and provide practical experience.

Tools and Technologies Used

During the internship, we used several tools and technologies, including:

- **XAMPP:** For local server setup and database management.
- **React JS:** For front-end development.
- **SQL:** For database interactions.
- **HTML, CSS, JavaScript, PHP:** For coding and developing web applications.

Mentorship and Guidance

The mentorship and guidance provided by the instructors were invaluable. Each participant was assigned a mentor who provided personalized feedback and support. Regular review sessions ensured that we stayed on track and received timely assistance with any challenges we faced. The instructors' expertise and willingness to help greatly enhanced the learning experience.

3. TECHNICAL SKILLS ACQUIRED

XAMPP

Role in Web Development: XAMPP is an open-source platform that provides an easy-to-install Apache distribution containing MySQL, PHP, and Perl. It is a crucial tool for local development and testing of web applications before deploying them to a live server.

Setup Process: Setting up XAMPP involves downloading the installer from the official website, running the setup wizard, and configuring the necessary components. Once installed, the XAMPP control panel allows easy management of the Apache server and MySQL database.

Server Configuration: Configuring the server involves setting up virtual hosts, managing port numbers, and adjusting PHP settings. This setup is essential for creating a local development environment that closely mirrors a live server.

Database Management: XAMPP includes phpMyAdmin, a powerful tool for managing MySQL databases. Through phpMyAdmin, I learned to create databases, tables, and manage data using SQL queries. This hands-on experience was instrumental in understanding database interactions.

Coding

HTML: The backbone of web development, HTML is used to create the structure of web pages. I learned to use various HTML tags to create elements such as headings, paragraphs, lists, forms, and tables.

CSS: Cascading Style Sheets (CSS) are used to style web pages. I gained proficiency in using CSS to apply styles, create layouts, and enhance the visual appeal of web pages. This included learning about selectors, properties, and responsive design techniques.

JavaScript: JavaScript is a powerful scripting language used to create dynamic and interactive web content. I learned to use JavaScript for tasks such as form validation, DOM manipulation, and creating interactive features like sliders and modals.

PHP: PHP is a server-side scripting language used to create dynamic web pages. I learned to use PHP for tasks such as handling form submissions, interacting with databases, and creating dynamic content based on user input.

Login Page Development

Designing the Login Page: The process of designing a login page involved creating a userfriendly interface with HTML and CSS. This included designing form elements such as input fields, labels, and buttons.

Developing the Login Functionality: Developing the login functionality required integrating front-end and back-end components. This involved using PHP to handle form submissions, validate user credentials, and establish sessions.

Security Considerations: Ensuring the security of the login page was crucial. I learned about best practices for securing user data, such as hashing passwords, using prepared statements to prevent SQL injection, and implementing HTTPS.

React JS

Fundamentals of React JS: React JS is a JavaScript library for building user interfaces. I learned about its core concepts, including components, state management, and JSX.

Components: Components are the building blocks of React applications. I learned to create reusable components and manage their state and props.

State Management: State management is crucial for managing dynamic data in React applications. I learned to use React's built-in state management and context API to handle complex state interactions.

JSX: JSX is a syntax extension for JavaScript that allows writing HTML-like code within JavaScript. I learned to use JSX to create dynamic and interactive user interfaces.

SQL

Data Types: SQL supports various data types such as integers, strings, dates, and more. I learned to choose appropriate data types for different fields in a database.

Queries: SQL queries are used to interact with databases. I learned to write queries for tasks such as inserting, updating, and deleting data, as well as retrieving data using SELECT statements.

Joins: Joins are used to combine data from multiple tables. I learned about different types of joins, such as INNER JOIN, LEFT JOIN, and RIGHT JOIN, and how to use them to retrieve related data from multiple tables.

4. PROJECTS AND ASSIGNMENTS

Detailed Description of the Employee Data Insert Project

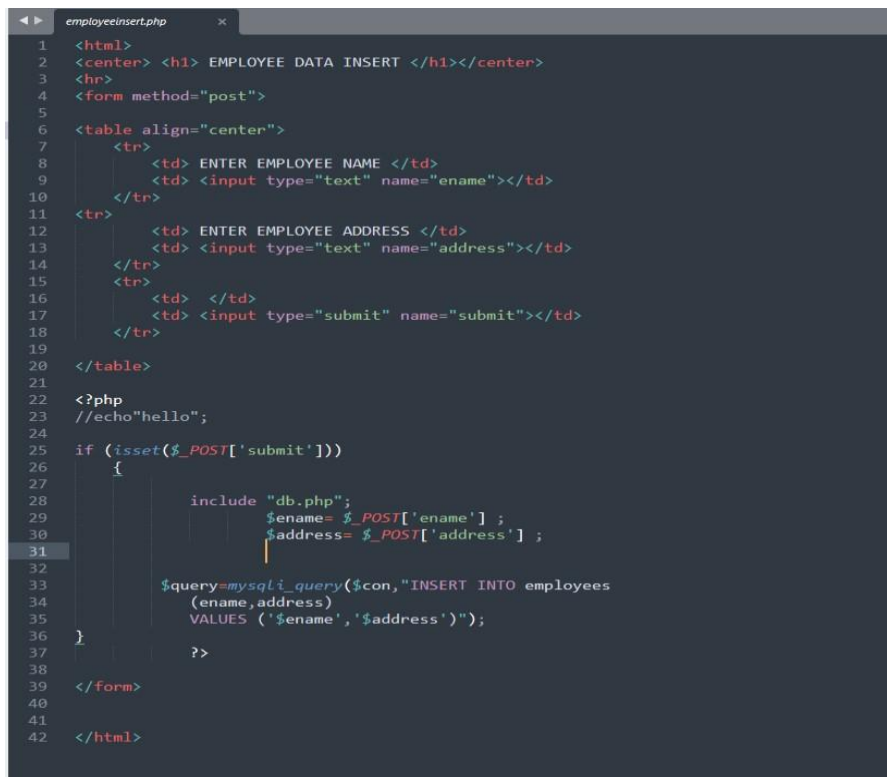
Problem Statement: The project involved creating a simple web application to insert employee data into a database. The application required a form for entering employee name and address, and a script to handle the form submission and insert the data into a MySQL database.

Solution Approach: The solution was developed using HTML, CSS, PHP, and MySQL. The front-end form was designed using HTML and CSS, while PHP was used for handling the form submission and interacting with the MySQL database.

Outcomes: The completed application allowed users to enter employee data through a web form and store the data in a MySQL database. This project demonstrated my ability to create a simple yet functional web application that integrates front-end and back-end technologies.

Role and Contributions: I was responsible for designing the form, writing the PHP script to handle form submissions, and configuring the MySQL database.

Screenshots and Code Snippets:



```
1 <html>
2 <center> <h1> EMPLOYEE DATA INSERT </h1></center>
3 <hr>
4 <form method="post">
5
6 <table align="center">
7
8     <tr>
9         <td> ENTER EMPLOYEE NAME </td>
10        <td> <input type="text" name="ename"></td>
11    </tr>
12
13    <tr>
14        <td> ENTER EMPLOYEE ADDRESS </td>
15        <td> <input type="text" name="address"></td>
16    </tr>
17
18    <tr>
19        <td> </td>
20        <td> <input type="submit" name="submit"></td>
21    </tr>
22 </table>
23
24 <?php
25 //echo"hello";
26
27 if (isset($_POST['submit']))
28 {
29     include "db.php";
30     $ename= $_POST['ename'] ;
31     $address= $_POST['address'] ;
32
33     $query=mysqli_query($con,"INSERT INTO employees
34     (ename,address)
35     VALUES ('$ename','$address')");
36 }
37
38 ?>
39 </form>
40
41
42 </html>
```

Detailed Description of the My Profile Project

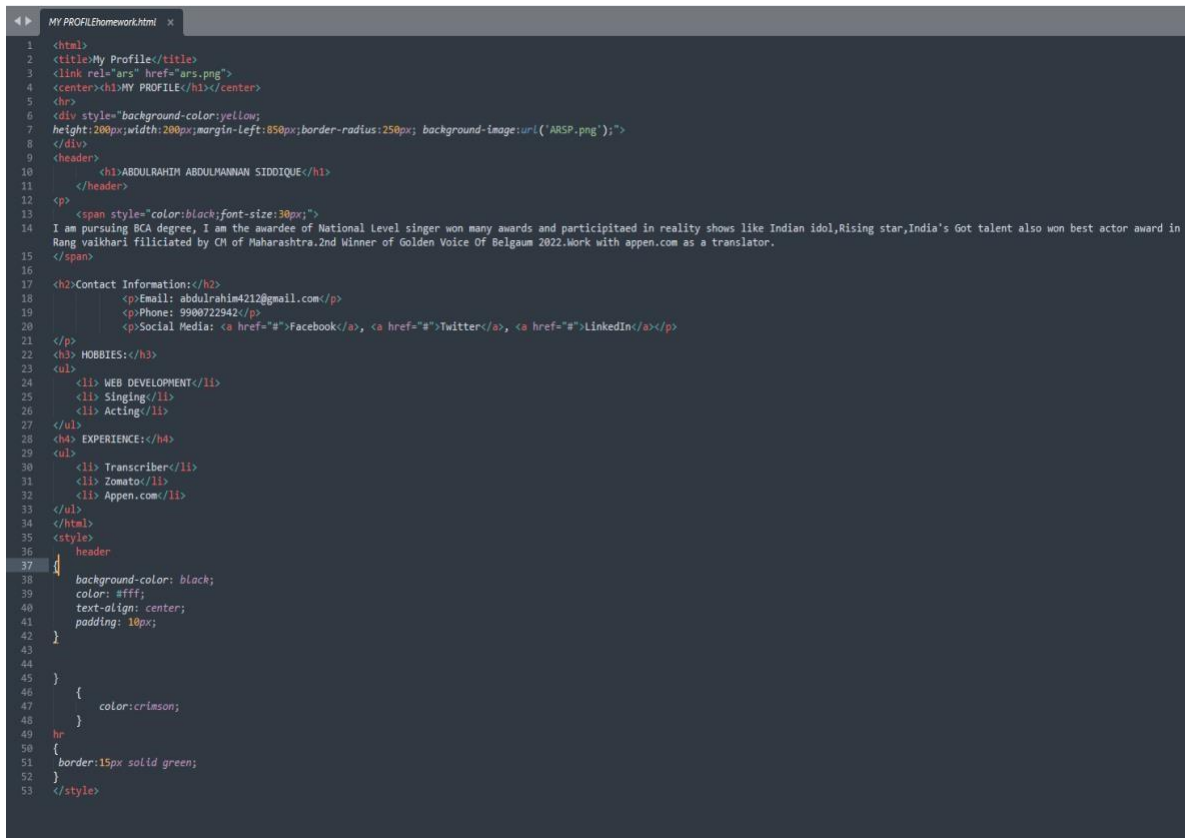
Problem Statement: The project involved creating a personal profile page that showcases personal information, achievements, contact details, hobbies, and work experience. The page was designed to be visually appealing and provide a comprehensive overview of the individual's background.

Solution Approach: The solution was developed using HTML and CSS to create the structure and style of the profile page. The page included various elements such as headings, paragraphs, lists, and images to display the information in a structured manner.

Outcomes: The completed profile page provided a detailed overview of the individual's personal and professional background. This project demonstrated my ability to design and develop a static web page using HTML and CSS.

Role and Contributions: I was responsible for designing the layout of the page, writing the HTML code to structure the content, and using CSS to style the elements.

Screenshots and Code Snippets:



```
1 <html>
2 <title>My Profile</title>
3 <link rel="ars" href="ars.png">
4 <center><h1>MY PROFILE</h1></center>
5 <hr>
6 <div style="background-color:yellow;
7 height:200px;width:200px;margin-left:850px;border-radius:250px; background-image:url('ARSP.png');">
8 </div>
9 <header>
10 <h1>ABDULRAHIM ABDULMANNAN SIDDIQUE</h1>
11 </header>
12 <p>
13 <span style="color:black;font-size:30px;">
14 I am pursuing BCA degree, I am the awardee of National level singer won many awards and participated in reality shows like Indian idol,Rising star,India's Got talent also won best actor award in
15 Rang varkharil filicited by CM of Maharashtra.2nd Winner of Golden Voice Of Belgaum 2022.Work with appen.com as a translator.
16 </span>
17 <h2>Contact Information:</h2>
18 <p>Email: abdulrahim4212@gmail.com</p>
19 <p>Phone: 9900722942</p>
20 <p>Social Media: <a href="#">Facebook</a>, <a href="#">Twitter</a>, <a href="#">LinkedIn</a></p>
21 </p>
22 <h3>HOBBIES:</h3>
23 <ul>
24 <li>WEB DEVELOPMENT</li>
25 <li>Singing</li>
26 <li>Acting</li>
27 </ul>
28 <h4>EXPERIENCE:</h4>
29 <ul>
30 <li>Transcriber</li>
31 <li>Zomato</li>
32 <li>Appen.com</li>
33 </ul>
34 </html>
35 <style>
36 <header>
37 {
38 background-color: black;
39 color: #fff;
40 text-align: center;
41 padding: 10px;
42 }
43 }
44 }
45 {
46 color:red;
47 }
48 }
49 <hr>
50 {
51 border:15px solid green;
52 }
53 </style>
```

5. Challenges and Learnings

Challenges Faced and Overcoming Strategies

Challenge 1: Debugging Code: One of the major challenges was debugging complex code, especially when integrating multiple technologies. I overcame this by systematically isolating the problem areas and using debugging tools and techniques.

Challenge 2: Time Management: Balancing multiple projects and assignments required effective time management. I developed a schedule to allocate specific times for different tasks, ensuring that all deadlines were met.

Challenge 3: Learning New Technologies: Learning new technologies such as React JS and SQL within a limited time frame was challenging. I utilized online resources, documentation, and peer discussions to accelerate my learning process.

Key Learnings and Insights

Technical Skills: The internship provided an opportunity to enhance my technical skills, particularly in full stack web development. I gained hands-on experience with tools and technologies that are highly relevant in the industry.

Problem-Solving Skills: Working on real-world projects helped me develop problem-solving skills. I learned to approach problems methodically, break them down into smaller tasks, and find effective solutions.

Collaboration: The importance of teamwork and collaboration was a key takeaway. Working with peers and mentors provided different perspectives and insights, leading to better project outcomes.

Importance of Teamwork and Collaboration

Collaboration was essential in completing the projects successfully. Regular team meetings, code reviews, and brainstorming sessions fostered a collaborative environment. Sharing knowledge and helping each other overcome challenges enhanced the overall learning experience.

6. CONCLUSION

Summary of Internship Experience

The internship at Edureka was a highly enriching experience that allowed me to immerse myself in the world of Full Stack Web Development and React JS. Over the course of the internship, I was introduced to a variety of tools, technologies, and best practices that are crucial for a successful career in web development.

The structured program provided a comprehensive understanding of both front-end and backend development. I had the opportunity to work on real-world projects, which not only enhanced my technical skills but also helped me gain practical experience in applying theoretical concepts. The hands-on approach facilitated a deeper understanding of how different technologies interact and how to effectively integrate them to build functional web applications.

Acquired Skills and Their Relevance

Throughout the internship, I acquired a range of technical skills that are directly applicable to my future career goals:

- **XAMPP:** I learned how to set up and manage a local server environment using XAMPP, which is essential for developing and testing web applications before deploying them to a live server.
- **HTML, CSS, and JavaScript:** My knowledge of core web technologies was reinforced, and I gained expertise in creating responsive and interactive web pages.
- **PHP and SQL:** I developed a strong foundation in server-side scripting with PHP and database management with SQL, which are crucial for building dynamic web applications.
- **React JS:** I acquired hands-on experience with React JS, learning how to create component-based user interfaces and manage application state efficiently.

These skills are highly relevant to the current job market, where proficiency in both front-end and back-end technologies is often required. The ability to build and manage full-stack web applications will be invaluable as I pursue career opportunities in web development.

Reflection on Learning and Development

Reflecting on the internship, I realize how much I have grown both professionally and personally. The challenges I faced, such as debugging complex code and managing multiple projects, helped me develop resilience and problem-solving skills. I learned to approach tasks methodically, seek help when needed, and continuously improve my skills.

The exposure to new technologies and development practices broadened my understanding of the web development landscape. Additionally, the opportunity to work on collaborative projects taught me the importance of teamwork and effective communication in achieving project goals.

Future Career Goals

The skills and experiences gained during this internship have significantly contributed to my career development. I am now better equipped to pursue advanced roles in web development, such as full-stack developer or front-end engineer. The knowledge of modern web technologies, combined with practical project experience, positions me well for future opportunities in the tech industry.

I am particularly interested in exploring further into specialized areas such as advanced React JS techniques, cloud-based solutions, and emerging web technologies. The internship has also sparked a keen interest in contributing to open-source projects and continuous learning to stay updated with industry trends.

Expression of Gratitude

I am deeply grateful for the opportunity to intern at Edureka. The guidance and support provided by my mentors and instructors were instrumental in my learning journey. Their expertise, feedback, and encouragement were invaluable in helping me navigate through complex topics and projects.

I also appreciate the collaborative environment fostered by my peers. The teamwork and shared learning experiences enriched my internship and made it a memorable and productive experience.

In conclusion, the internship at Edureka has been a significant step in my professional development, and I am excited to apply the knowledge and skills acquired in my future endeavors.